RV COLLEGE OF ENGINEERING®
(Autonomous Institution Affiliated to VTU, Belagavi)
Bengaluru-560 059



CERTIFICATE

This is to certify that …….

bearing USN……..…has satisfactorily completed the course of experiments in the laboratory component of the course……..……

with the course code ……prescribed by the department of Chemical Engineering for…… .Semester of BE Program during the academic year…….

CIE Marks for Laboratory component:

Laboratory Exercises (Maximum 40)	
Test (Maximum 10)	
Total (50)	

Signature of Head of the Department *Signature of the faculty in-charge*

Date:

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EXPERIMENTS CONDUCTED

Sl. No.	Title of the Experiment	Page No.	Marks (40)	Signature

Assessment for each experiment:

Sl. No	Assessment Criteria	Percentage of marks	Marks
1	Preparedness	20%	8
2	Conduction of Experiment	40%	16
3	Results and discussions	20%	8
4	Viva Voce	20%	8

CIE for Laboratory component: Average of all experiments marks (40) + Test (10)

3.1 FLOW THROUGH OF VENTURIMETER

3.1.0 Nomenclature:

β	Ratio of throat diameter to pipe diameter	-
ΔH	Fluid head loss across Venturimeter	m
μ	Coefficient of viscosity	kg/ms
ρ	Density of water	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
A_T	Area of the throat	m^2
C_v	Coefficient of discharge	-
D_t	Diameter of the throat	m
D_P	Diameter of the pipe	m
g	Acceleration due to gravity	m/s^2
LHS	Left leg liquid level in manometer	m
RHS	Right leg liquid level in manometer	m
R_m	Difference in liquid levels in manometer	m
Q_a	Actual flow rate	$\frac{m^3}{s}$
Q_{th}	Theoretical flow rate	$\frac{m^3}{s}$
v_a	Actual fluid velocity	m/s
P	Pressure of the fluid	N/m^2
v	Average velocity of the fluid across the section of the pipe	m/s

3.1.1 Aim:

- To find the coefficient of discharge of Venturimeter
- To determine relationship between coefficient of discharge and Reynolds number
- To find the relationship between differential pressure and flow rate of the fluid.

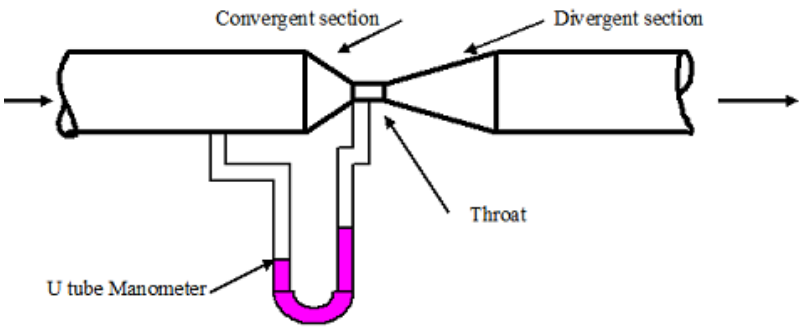
3.1.2 Theory:

Measurement of the flow rate of a fluid used in process is essential in all industries. One of the commonly used methods of flow measurement is to place a resistance in the pipe carrying the fluid and to measure the pressure drop across

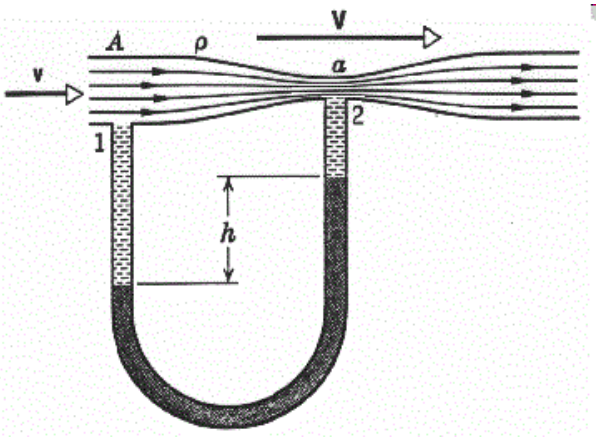
the resistance. A simple method of introducing a resistance is to restrict the area of cross section of flow by inserting an element in the line. One of such elements is a venture.

A Venturi consists of a flanged inlet section connected at either end to the pipeline by conical sections. The entrance cone angle is about $23°$ and the exit cone angle varies between $5° - 15°$. The two cones are connected by short length of pipe of constant cross section called throat. The throat length equals one throat diameter.

At constant flow rate, the velocity at the throat is greater than that in the pipe. The increase in velocity results in an increase in kinetic energy. As the total energy should be constant, there must be a corresponding reduction in pressure. Thus, as shown in the figure, the downstream pressure is always less than the upstream pressure. The difference in pressure corresponding to different flow rates are found by connecting a U-tube manometer across the venturi inlet and throat.



The expression for theoretical flow rate is obtained by applying the continuity equation and energy equation at inlet and throat section.



Applying Bernoulli's equation across point 1 and point 2, if v_1 and v_2 are the average velocities at point 1 and point 2 respectively then,

$$(\frac{P_1}{\rho g}) + (\frac{v_1^2}{2g}) + z_1 = (\frac{P_2}{\rho g}) + (\frac{v_2^2}{2g}) + z_2$$

$$(v_2^2 - v_1^2)/_{2g} = \left(P_1/_{\rho g}\right) - (P_2/_{\rho g})$$

Now applying the equation of continuity at both points, we have

$$A_1V_1 = A_2V_2v_2 = \frac{A_1v_1}{A_2}v_1 = d_2^2 \times v_2/d_1^2$$

Now substituting the value of v_1 in the above expression for kinetic head loss and if $\beta = d_2/d_1$, we have

$$v_2 = \sqrt{\frac{2(P_1 - P_2)}{\rho(1-\beta^4)}}$$

We have $Q = A_2v_2$.

$$Q_{th} = A_2\sqrt{\frac{2(P_1 - P_2)}{\rho(1-\beta^4)}}$$

Q_{th} is the theoretical flow rate as computed from above equation and applies to frictionless flow of incompressible fluids. Actual flow includes frictional loss between point 1 and 2

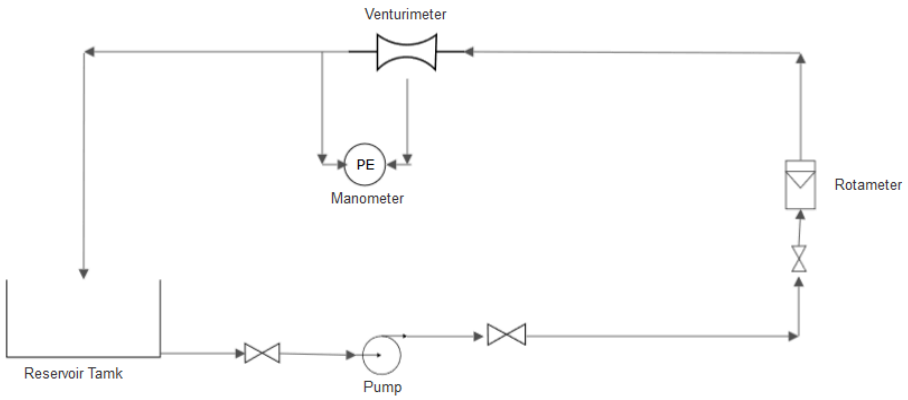
$$Q_{act} = C_DQ_{th} = C_DA_2\sqrt{\frac{2(P_1 - P_2)}{\rho(1-\beta^4)}}$$

where C_D is called as co-efficient of discharge which accounts for small friction between points 1 and 2 and depends upon the type of flow, type of fluid and dimensions of venturi tube and pipe. For a well-designed venturimeter the constant C_D is about 0.98 for pipe diameters of 2 to 8 inches and about 0.99 for larger sizes.

The equation relating flow rate to pressure drop is

$$Q = C_DA_t\sqrt{\frac{2(P_1 - P_2)}{\sqrt{(1-\beta^4)\rho}}}$$

3.1.3 Experimental Set up / Line diagram:



3.1.4 Procedure:

- Check the connections as per the line diagram.
- Check manometer levels under no flow conditions.
- Remove any entrapped air bubbles from the manometer.
- Allow the water to flow through venturimeter and regulate the flow with the help of gate valve.
- Once the manometer levels are steady, note the difference in level of manometer fluid.
- Measure the flow rate.
- Repeat the procedure for various flow rates.
- Determine volumetric flow rate, coefficient of discharge and Reynolds number.
- Plot a graph of Q_a v/s ΔH on an ordinary graph and C_v v/s N_{Re} on a semi log graph.

3.1.5 Observations:

Trail No.	Manometer Reading		Flow Rate	
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				

4					
5					
6					
7					
8					
9					
10					

3.1.6 **Specifications:**

Diameter of pipe $D_p = 0.0254\ m$

Diameter of throat $D_T = 0.0125\ m$

Density of Manometric fluid at room temperature $\rho_m =$ _____ kg/m^3

Density of water at room temperature $\rho =$ _____ kg/m^3

Formulae:

Fluid head loss between the pipe and throat:

$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Actual flow rate:

$Q_a = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$

Actual velocity:

$$A_T = \frac{\pi(D_T)^2}{4}\ (m^2)$$

$v_a = \frac{Q_a}{A_T}\ \left(\frac{m}{s}\right)$

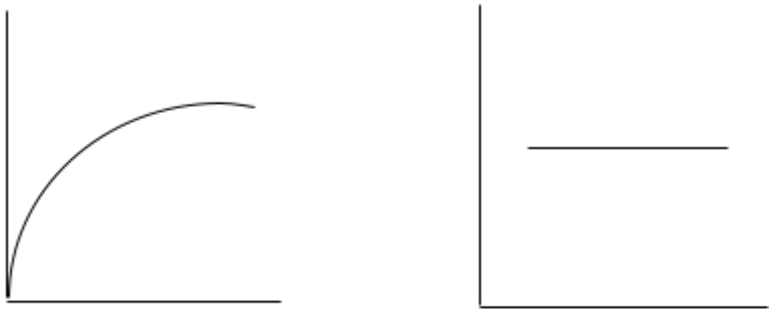
Theoretical flow rate :

$$\beta = \frac{D_T}{D_P}$$

$Q_{th} = A_T\sqrt{\frac{2g\Delta H}{1-\beta^4}}\ \left(\frac{m^3}{s}\right)$

Reynolds Number:

$N_{Re} = \frac{D_T v_a \rho}{\mu}$

	Co-efficient of Discharge: $C_v = \frac{Q_a}{Q_{th}}$ Graphs: 
3.1.7	**Specimen Calculations:**

3.1.8	**Results and discussions**
	3.1.8.1 **Results:**

Trial No	R_m (m)	ΔH (m)	$Q_a\left(\frac{m^3}{s}\right)$	$v_a\left(\frac{m}{s}\right)$	$Q_{th}\left(\frac{m^3}{s}\right)$	C_v	N_{Re}
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

3.1.8.2	**Discussion:**

3.1.9	**Conclusions:**	
3.1.10	**Assessment:**	

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
		Total	40	

Signature of the student **Signature of the faculty with date**

3.2 FLOW THROUGH ORIFICE METER

3.2.0 Nomenclature:

β	Ratio of orifice diameter to pipe diameter	-
ΔH	Fluid head loss across Venturimeter	m
μ	Coefficient of viscosity	kg/ms
ρ	Density of water	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
A_o	Area of the orifice	m^2
C_d	Coefficient of discharge	-
D_o	Diameter of the orifice	m
D_P	Diameter of the pipe	m
g	Acceleration due to gravity	m/s^2
LHS	Left leg liquid level in manometer	m
RHS	Right leg liquid level in manometer	m
R_m	Difference in liquid levels in manometer	m
v_a	Actual fluid velocity	m/s
Q_a	Actual flow rate	$\frac{m^3}{s}$
Q_{th}	Theoretical flow rate	$\frac{m^3}{s}$

3.2.1 Aim:

- To find the coefficient of discharge of orifice meter
- To determine relationship between coefficient of discharge and Reynolds Number
- To find the relationship between differential pressure across the orifice and flow rate of the fluid.

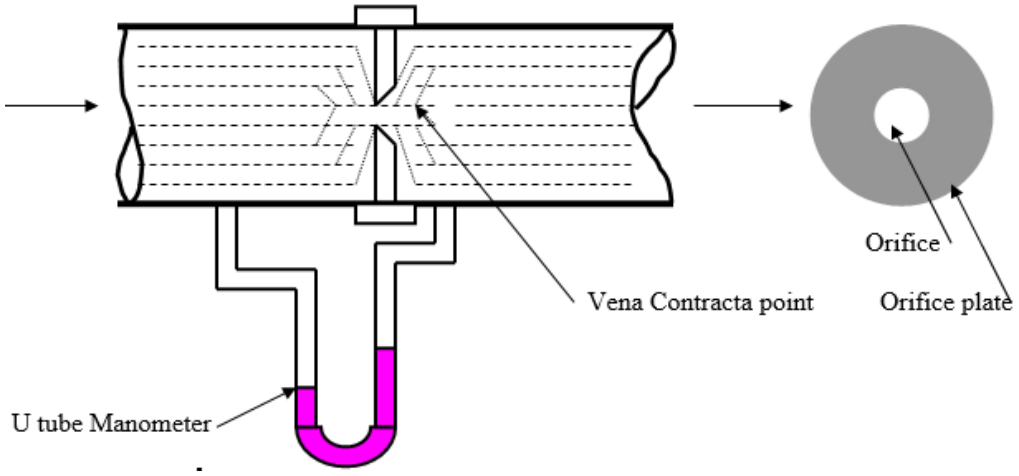
3.2.2 Theory:

An orifice meter is a differential pressure flow meter which reduces the flow area using an orifice plate. An orifice is a flat plate with a centrally drilled hole machined to a sharp edge. The orifice plate is inserted between two flanges perpendicularly to the flow, so that the flow passes through the hole with the sharp edge of the orifice pointing to the upstream. The relationship between flow rate and pressure drop can be determined using Bernoulli's equation as:

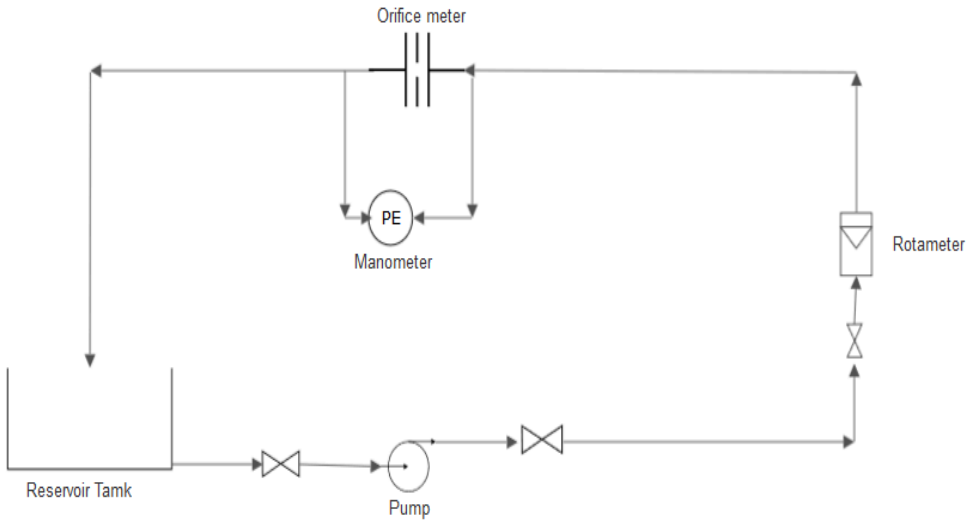
$$Q = C_d A_o \sqrt{\frac{2(P_1 - P_2)}{(1-\beta^4)\rho}}$$

where, Q is the volumetric flow rate, A_o is the orifice cross sectional area,p_1 and p_2 are the pressure measured at the upstream and downstream of the orifice and C_d is the discharge coefficient for the orifice β is the is the ratio of orifice diameter to the pipe diameter.

As the fluid flows through the orifice plate the velocity increases, at the expense of pressure head. The pressure drops suddenly as the orifice is passed. It continues to drop until the vena contracta is reached and then gradually increases until at approximately 5 to 8 diameters downstream where a maximum pressure point is reached.



3.2.3 Experimental Set up / Line diagram:



3.2.4 Procedure:

- Check the connections as per the line diagram.
- Check manometer levels under no flow conditions.

- Remove any entrapped air bubbles from the manometer
- Allow the water to flow through orifice meter and regulate the flow with gate valve.
- Once the manometer levels are steady, note the difference in levels of manometer fluid.
- Measure the flow rate.
- Repeat the procedure for various flow rates.
- Calculate volumetric flow rate, coefficient of discharge and Reynolds number.
- Draw a graph of Q_a v/s ΔH on an ordinary graph and C_d v/s N_{Re} on a semi log graph.

3.2.5 Observations:

Trail No.	Manometer Reading		Flow Rate	
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

3.2.6 **Formulae:**

Specifications:

Diameter of pipe $D_p = 0.0254\ m$

Diameter of the orifice $D_o = 0.0125\ m$

Density of Manometric fluid at room temperature $\rho_m =$ _____ kg/m^3

Density of water at room temperature $\rho =$ _____ kg/m^3

Fluid head loss across the Venturi:

$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Actual flow rate:

$Q_a = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$

Actual velocity:

$$A_o = \frac{\pi(D_o)^2}{4}\ (m^2)$$

$v_a = \frac{Q_a}{A_o}\ \left(\frac{m}{s}\right)$

Theoretical flow rate :

$$\beta = D_o/D_p$$

$Q_{th} = A_o\sqrt{\frac{2g\Delta H}{1-\beta^4}}\ \left(\frac{m^3}{s}\right)$

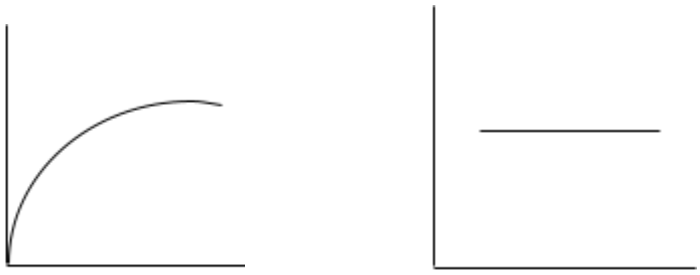
Reynolds Number:

$N_{Re} = \frac{D_o v_a \rho}{\mu}$

Co-efficient of Discharge:

$C_d = \frac{Q_a}{Q_{th}}$

Graphs:



3.2.7 **Specimen Calculations:**

3.2.8	**Results and discussions**

3.2.8.1 Results:

Trial No	R_m (m)	ΔH (m)	$Q_a\left(\frac{m^3}{s}\right)$	$v_a\left(\frac{m}{s}\right)$	$Q_{th}\left(\frac{m^3}{s}\right)$	C_d	N_{Re}
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

3.2.8.2 Discussion:

3.2.9	**Conclusions:**	
3.2.10	**Assessment:**	

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	

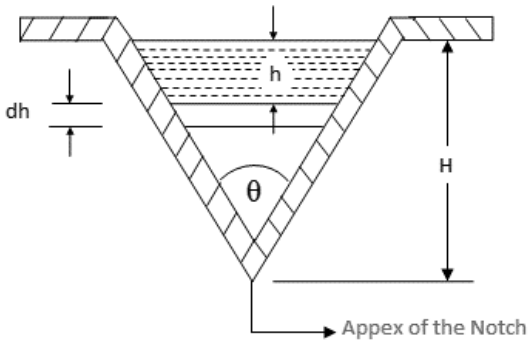
		2	Viva Voce	20%	8	
		3	Conduction of Experiment	40%	16	
		4	Results and discussions	20%	8	
			Total		40	

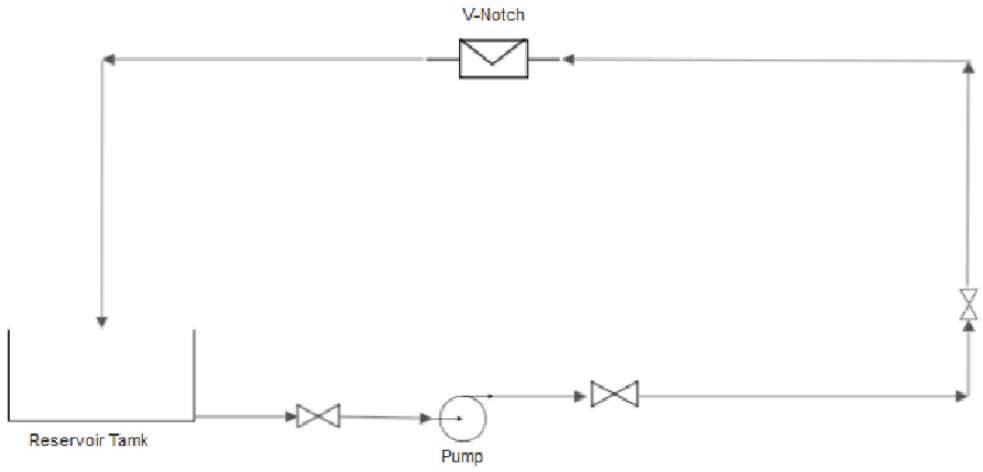
Signature of the student **Signature of the faculty with date**

3.3 FLOW THROUGH V-NOTCH

3.3.0 Nomenclature:

α	Angle of inclination of the scale used to measure the level of water in the tank	-
θ	Apex angle of the V-notch	-
A_t	Cross sectional area of collection tank	m^2
C_d	Coefficient of discharge	-
H	Fluid head	m
L	Height of water above the notch crest	m
L_1	Height of water in the tank when water level is just at the notch apex	m
L_2	Height of water in the tank at steady state for a flow rate of water into the tank	m
L_t	Height of water collected in the collection tank in time $t\ s$	m
Q_a	Actual flow rate of water	m^3/s
Q_t	Theoretical flow rate of water	m^3/s
t	Time for which water is collected to measure flow rate	s

3.3.1	**Aim:** ● To determine Notch constants ● To find coefficient of discharge of the V-Notch.
3.3.2	**Theory:** A notch is an opening in the side of a measuring tank or reservoir extending above the free surface. These notches are used to measure discharge of open channel flows, by passing or placing or constructing them across the stream. Notches are generally used for measuring discharge in small open channels or laboratory flumes. Notches can be of different shapes such as triangular, rectangular, trapezoidal, stepped notch, etc. the bottom of the notch over which the water flows is known as crest or sill and the thin sheet of water flowing through the notch is known as nappe or vein. The edges of the notch are beveled on the downstream side so as to have a sharp-edged sides and crest resulting in minimum contact with the flowing fluid. The discharge over notch is measured by measuring the head acting over the notch. The V-notch or triangular notch is sharp crested notch, which is mainly used to determine the low rate of flow.  **Fig : Triangular Notch** $Q = \frac{8}{15} C_d \sqrt{2g} \times \tan tan\left(\frac{\theta}{2}\right) \times H^{\frac{5}{2}}$ This is the relationship between flow rate and head measured at the notch.
3.3.3	**Experimental Set up / Line diagram:**

	
3.3.4	**Procedure:** ● Allow the water to flow into the tank till the level of the water is just at the Notch apex. ● Note the level of the water in the tank using the inclined scale. This will be referred to as initial height. ● Allow the water to flow over the v-notch and regulate the flow with gate valve. ● Once the level is steady, note height of liquid above the notch apex using inclined scale. ● Measure the flow rate of the tank by measuring volume of water collected in the reservoir tank for a known time.

3.3.5

Trial No	Inclined scale reading		Measurement of flow rate	
	Initial Height, L_1(cm)	Final Height, L_2(cm)	Increase in level of water in reservoir tank, L_t (cm)	Time (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Observations:

3.3.6 **Formulae:**

Specifications:

Angle of V- Notch $(\theta) = 62°$

Angle of inclination of the scale$(\alpha) = 12°$

Cross sectional area of the collection tank $A_t =$ _____ m^2

Head of the water developed above the notch crest: $L = L_2 - L_1\ (m)$

Fluid Head: $H = L\,Sin\alpha\ (m)$

Actual Volumetric Flow Rate:

$$Q_a = \frac{L_t \times A_t}{t} \quad \left(\frac{m^3}{s}\right)$$

Theoretical Volumetric Flow Rate:

$$Q_t = \frac{8}{15}\sqrt{2g}\tan tan\left(\frac{\theta}{2}\right)H^{\frac{5}{2}} \quad \left(\frac{m^3}{s}\right)$$

Coefficient of discharge:

	$$C_d = \frac{Q_a}{Q_t}$$ Determination of coefficient of discharge using graphical method: $$Q = \frac{8}{15}\sqrt{2g}\tan tan\left(\frac{\theta}{2}\right)H^{\frac{5}{2}}\ Q = kH^{\frac{5}{2}}k = C_d\left(\frac{8}{15}\right)\sqrt{2g}\tan tan\left(\frac{\theta}{2}\right)$$ Plot a graph of $Q_a \ \ v/s \ H$ on a log-log graph. Compare the slope of the line with the power of H in the theoretical flow rate equation. Obtain y-intercept and determine the value of k. $$C_d(graphically) = \frac{k}{\left(\frac{8}{15}\right)\sqrt{2g}}$$
3.3.7	**Specimen Calculations:**

3.3.8	**Results and discussions**
	3.2.8.1 **Results:**

Trial No	$H(m)$	$Q_a\left(m^3/s\right)$	$Q_{th}\left(m^3/s\right)$	C_d
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

3.3.8.2	**Discussion:**
3.3.9	**Conclusions:**

3.3.10	**Assessment:**

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**

3.4 FLOW THROUGH OPEN ORIFICE

3.4.0	**Nomenclature:**

	A_o	Area of orifice	m^2
	A_c	Area of vena-Contracta	m^2
	C_c	Coefficient of contraction	-
	C_d	Coefficient of discharge	-
	C_v	Coefficient of velocity	-
	H	Water level in orifice tank	m
	H_1	Initial height of water in the tank before emptying.	m
	H_2	Final height after the tank was allowed to drain.	m
	L_t	Level of water in the collection tank	m
	Q_a	Actual flow rate	m^3/s
	Q_{th}	Theoretical flow rate	m^3/s
	T	Actual time taken for level to change from H_1 to H_2 without inlet flow	s
	v_a	Actual velocity	m/s
	v_{th}	Theoretical velocity	m/s
	X	Horizontal length of fluid jet	m
	Y	Vertical height of fluid jet	m

3.4.1 **Aim:**

- To determine the hydraulic coefficients; coefficient of velocity, coefficient of discharge and coefficient of contraction of an open orifice.
- To calculate actual time taken for change in liquid level of tank due to the flow through the orifice when there is no inflow in the tank and compare with the theoretical time.

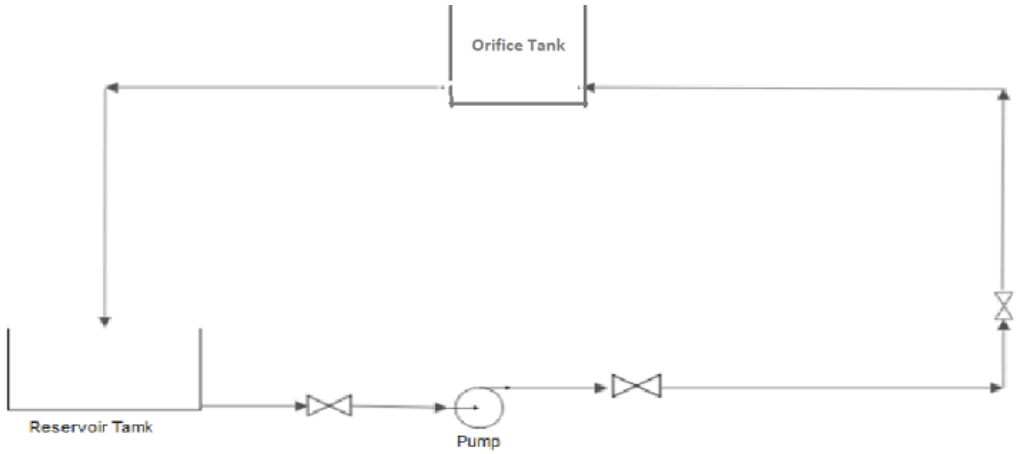
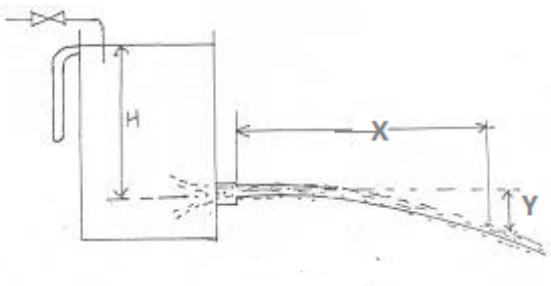
3.4.2 **Theory:**

An open-orifice is a circular opening in the side of an open tank, with closed perimeter in contact with the liquid which flows through it. The hydraulic co-efficient of an orifice are

- Co-efficient of contraction.
- Co-efficient of velocity.
- Co-efficient of discharge.

Coefficient of contraction is defined as the ratio of the area at vena contraction to that of the orifice.

$C_c = A_c/A_o$

	The value of C_c depends on a number of factors such as, shape of the orifice, size of the orifice, head on the orifice and viscosity of the liquid. Coefficient of velocity is defined as the ratio of the actual velocity of the jet at vena Contracta to the theoretical velocity of flow through the orifice. $C_v = v_a/v_{th}$ The actual velocity of flow is less than the theoretical due to the loss of frictional energy and the curvature of the streamlines. Coefficient of discharge is defined as the ratio of the actual discharge to the theoretical discharge through the orifice. $C_d = Q_a/Q_{th}$ $C_d = C_v C_c$ C_d lies in the range of $0.61 - 0.64$ C_v lies in the range of $0.95 - 0.99$ C_c lies in the range of $0.5 - 1.0$
3.4.3	**Experimental Set up / Line diagram:**  
3.4.4	**Procedure:**

- Allow the water into the rectangular tank and regulate the flow with gate valve.
- Maintain a constant level of water by using the drain valve at bottom.
- Record the following
  - The horizontal length of a point in fluid jet.
  - The vertical height of fluid jet.
  - Height of water in tank above the orifice
  - Volume of water collected in known time.
- Repeat the above at least five trials starting from minimum to maximum flow rate.
- To find the time taken for emptying the tank, close the inlet valve and record the time taken for the fall in level of water.

3.4.5 **Observations:**

Trial No	Water level, $H(m)$	X(m)	$Y(m)$	Measurement of flow rate	
				Increase in level of water in reservoir tank, L_t (cm)	Time (s)
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Time taken to empty the tank:

Initial Height $H_1 =$ _____m

Final Height $H_2 =$ _____m

Actual time for change in level without inlet flow $T =$ _____s

3.4.6	**Formulae:** Specifications: Orifice diameter = $0.01\ m$ Area of the collection tank (A_t) = $0.033\ m^2$ Formulae: $Q_a = \frac{L_t \times A_t}{time}\ \left(\frac{m^3}{s}\right)$ $Q_{th} = A_o \times \sqrt{2gH}\ (\frac{m^3}{s})$ $C_d = Q_a / Q_{th}$ $C_v = \sqrt{(\frac{X^2}{4YH})}$ $C_c = C_d / C_v$ Theoretical time for the level to change (without inlet flow) from H_1 to H_2 : $$T_{th} = \frac{2A_t\left(\sqrt{H_1} - \sqrt{H_2}\right)}{C_d A_o \sqrt{2g}}$$
3.4.7	**Specimen Calculations:**

3.4.8 Results and discussions

3.4.8.1 Results:

Trial No	$Q_a\left(m^3/s\right)$	$Q_{th}\left(m^3/s\right)$	C_d	C_v	C_c
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

3.4.8.2 Discussion:

3.4.9 **Conclusions:**

3.4.10 **Assessment:**

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**

3.5 FLOW THROUGH STRAIGHT PIPE

3.5.0 Nomenclature:

μ	Viscosity of fluid flowing through the pipe	Ns/m^2
ρ	Density of fluid flowing through the pipe	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
A_p	Cross sectional area of the pipe	m^2
D_p	Diameter of the pipe	m
f	Friction factor	-
f_{th}	Theoretical friction factor	-
ΔH	Differential head across the pipe	m
L	Length of the pipe	m
N_{Re}	Reynolds Number	-
R_m	Difference in manometer leg levels	m
Q	Volumetric flow rate	m^3/s
v	Average velocity of fluid through the pipe	m/s

3.5.1 Aim:

- To find the variation of friction with respect to Reynolds number.
- To compare the experimentally calculated friction factor values with the theoretical values.
- To study the variation of friction with respect to Reynolds number for both experimental and theoretical values.

3.5.2 Theory:

The flow is

- laminar when $N_{Re} < 2100$
- turbulent transient when $2100 < N_{Re} < 4000$
- transient turbulent when $4000 < N_{Re}$

Correlations were made between the fanning friction factor and the Reynolds number for both laminar and turbulent flow in a variety of pipes.

For laminar flow, first principles can be used to develop a relation between Reynolds number and friction factor. The Hagen-Poiseuille equation relates the frictional pressure drop to fluid velocity, viscosity, and pipe dimension:

$$-(\Delta P)_f = \frac{32\mu VL}{D^2}$$

	Equating the pressure drop due to friction in the Hagen-Poiseuille equation, with the overall pressure drop across the pipe and combining with the Fanning results in relation for laminar flow as $$f = 16/N_{Re}$$ A first principle analysis cannot be used to develop a relation between friction factor and Reynolds numbers for turbulent flow. A variety of empirical correlations exist for turbulent flow in smooth pipes. Blasius equations: $$f = \frac{0.079}{N_{Re}^{0.25}}$$ The above correlation is valid for $4000 \leq N_{re} \leq 10^5$. The entire turbulent region can be represented by the von Kármán-Nikuradse equation as $\frac{1}{\sqrt{f}} = 4 \log log\, N_{Re}\sqrt{f} - 0.4$ for $N_{Re} > 4000$
3.5.3	**Experimental Set up / Line diagram:** *(Students to observe the set-up and draw the line/piping diagram)*

3.5.4 Procedure:

- Remove any entrapped air bubbles from the manometer.
- Allow the water to flow through the straight pipe regulating the flow with gate valve.
- Once manometer readings are steady, note the manometer readings.
- Collect the water for a known amount of time for the adjusted flow rate.
- Repeat the procedure for different flow rates.

3.5.5 Observations:

Trial No.	Manometer Reading		Flow Rate	
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

3.5.6 Formulae:

Specifications:

Length of the pipe $L = 1.6\,m$

Diameter of the pipe, $D_p = 0.0254\,m$

Density of CCl_4 at room temperature, ρ_m = _____ kg/m^3

Density of Water at room temperature, ρ = _____ kg/m^3

Viscosity of water, μ = _____ Ns/m^2

Formulae:

Fluid head loss across the pipe:

$$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$$

Flow rate:

$$Q = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$$

Velocity:

$$A_P = \frac{\pi\left(D_p\right)^2}{4} \quad (m^2)$$

$$v = \frac{Q}{A_p} \quad \left(\frac{m}{s}\right)$$

Reynolds Number:

$$N_{Re} = \frac{D_p v_a \rho}{\mu}$$

Fanning friction factor:

$$f = \frac{2gD_p\Delta H}{4Lv_a^2}$$

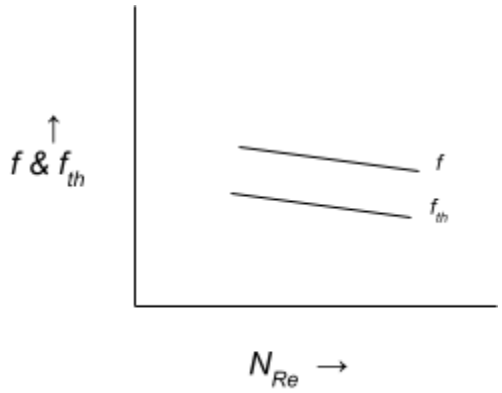
Theoretical value of f:

$$f_{th} = \frac{16}{N_{Re}} for\ laminar\ flow,\ N_{Re} < 2100$$

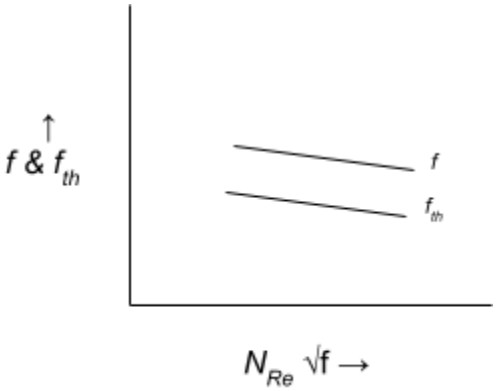
$$f_{th} = 0.079N_{Re}^{-0.25}\ for\ turbulent\ flow\ ,\ N_{Re} > 4000$$

Graphs:

a) $f\ Vs.\ NRe\ and\ f_{th}\ Vs\ N_{Re}\ (log\ -\ log\ graph)$



b) $f\ and\ f_{th}\ Vs.\ N_{Re} f\ (log\ -\ log\ graph)$



3.5.7	**Specimen Calculations:**

3.5.8	**Results and discussions**
	3.5.8.1 **Results:**

Trial No	R_m (m)	ΔH (m)	$Q(\frac{m^3}{s})$	$v\ (\frac{m}{s})$	N_{Re}	f	$N_{Re}\sqrt{f}$	f_{th}
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								

	3.5.8.2	**Discussion:**

3.5.9	**Conclusions:**	
3.5.10	**Assessment:**	

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**

3.6 FLOW THROUGH ANNULUS

3.6.0 Nomenclature:

μ	Viscosity of fluid flowing through the pipe	Ns/m^2
ρ	Density of fluid flowing through the pipe	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
A_a	Cross sectional area of the annulus	m^2
D_i	Inner diameter of the outer pipe	m
d_o	Outer diameter of the inner pipe	m
f	Friction factor	-
f_{th}	Theoretical friction factor	-
ΔH	Differential head across the pipe	m
L	Length of the pipe	m
N_{Re}	Reynolds Number	-
R_m	Difference in manometer leg levels	m
r_H	Hydraulic radius	m
Q	Volumetric flow rate	m^3/s
v	Velocity of fluid through the pipe	m/s

3.6.1 Aim:

- To find the variation of friction with respect to Reynolds number.
- To compare the experimentally calculated friction factor values with the theoretical values.
- To study the variation of friction with respect to Reynolds number for both experimental and theoretical values.

3.6.2 Theory:

An annulus is a closed channel between two concentric pipes. When a fluid is flowing through an annulus, there exists a frictional resistance because of viscous drag. This frictional resistance depends on the velocity of flow, the area of wetted surface and the nature of the surface.

Double pipe heat exchangers are used in industries for heat interchange between a heat & a cold fluid. There will be a pressure drop due to friction as a fluid flows

	through the inner pipe & also as the other fluid flows through the annulus. Fanning's equation is used to determine the pressure drop due to friction. $$\Delta H = \frac{\Delta P}{\rho g} = \frac{4fLv^2}{2Dg} \ (m)$$ In case of non- circular cross section, an equivalent diameter, D_e, is used. It is defined as 4(four) times the hydraulic radius r_H. Hydraulic radius is obtained by dividing the area of cross section by wetted perimeter. For a concentric pipe with inner pipe having outside diameter d_o & outer pipe having an inside diameter D_i. The annular area of cross section = $\pi / 4 (D_i^2 - d_0^2)$ Wetted perimeter = $\pi (D_i + d_0)$ D_e for annulus = $4\frac{\left[\frac{\pi}{4}\left(D_i^2 - d_o^2\right)\right]}{\pi\left(D_i + d_o\right)}$ $= (D_i - d_o)$ Fanning's equation is modified by substituting D_e for D.
3.6.3	**Experimental Set up / Line diagram:** *(Students to observe the set-up and draw the line/piping diagram. Symbol list may be referred to prepare the line diagram)*
3.6.4	**Procedure:** ● Connect the manometer between two tapings of the pipe fittings. ● Remove any entrapped air bubbles from the manometer.

- Allow the water to flow through the annulus regulating the flow with gate valve.
- Once manometer readings are steady, note the manometer readings.
- Collect the water for a known amount of time for the adjusted flow rate.
- Repeat the procedure for different flow rates.

3.6.5 Observations:

Trial No.	Manometer Reading		Flow Rate	
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

3.6.6 **Formulae:**

Specifications:

Length of the annular pipe $L = 2.6\,m$

Outer diameter of the inner pipe $d_o = 3.5\times10^{-2} m$

Inner diameter of the outer pipe, $D_i = 4.09\times10^{-2} m$

Density of CCl_4 at room temperature, ρ_m = _____ kg/m^3

Density of Water at room temperature, ρ = _____ kg/m^3

Viscosity of water, μ = _____ Ns/m^2

Formulae:

Fluid head loss across the pipe:

$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Flow rate:

$$Q = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$$

Annular flow area:

$$A_a = \frac{\pi}{4}\ (D_i^2 - d_o^2)\ \ (m^2)$$

Actual velocity :

$v_a = \frac{Q}{A_a}\ (\frac{m}{s})$

Calculation of equivalent diameter for annulus:

$$D_e = (D_i - d_o)$$

Reynolds Number:

$$N_{Re} = \frac{D_e v_a \rho}{\mu}$$

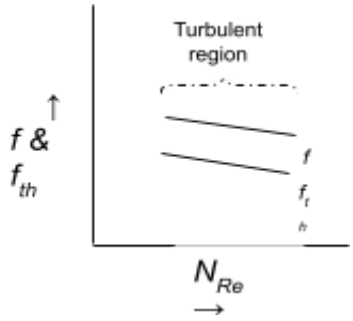
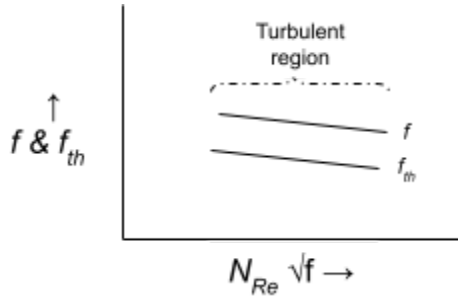
Fanning friction factor:

$$f = \frac{2gD_e \Delta H}{4Lv_a^2}$$

Theoretical value of f:

$f_{th} = \frac{16}{N_{Re}} for\ laminar\ flow,\ N_{Re} < 2100$

$f_{th} = 0.079 N_{Re}^{-0.25}\ for\ turbulent\ flow\ ,\ N_{Re} > 4000$

	Graphs: $f\ Vs.\ NRe\ and\ f_{th}\ Vs\ N_{Re}\ (log\ -\ log\ graph)$  $b)\ f\ and\ f_{th}\ Vs.\ N_{Re}f\ (log\ -\ log\ graph)$  Turbulent region f & f_{th} f f_{th} N_{Re} √f →

3.6.7	**Specimen Calculations:**

3.6.8	**Results and discussions**	
	3.6.8.1	**Results:**

Trial No	R_m (m)	ΔH (m)	$Q(\frac{m^3}{s})$	$v\ (\frac{m}{s})$	N_{Re}	f	$N_{Re}\sqrt{f}$	f_{th}
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								

	3.6.8.2	**Discussion:**
3.6.9	**Conclusions:**	

3.6.10 **Assessment:**

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**

3.7 FLOW THROUGH NON-CIRCULAR DUCTS

3.7.0 Nomenclature:

μ	Viscosity of fluid flowing through the pipe	Ns/m^2
ρ	Density of fluid flowing through the pipe	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
A_p	Cross sectional area of the pipe	m^2
D_p	Diameter of the pipe	m
f	Friction factor	-
f_{th}	Theoretical friction factor	-
ΔH	Differential head across the pipe	m
L	Length of the pipe	m
N_{Re}	Reynolds Number	-
R_m	Difference in manometer leg levels	m
Q	Volumetric flow rate	m^3/s
v	Velocity of fluid through the pipe	m/s

3.7.1 Aim:

To calculate friction factor for flow through ducts with triangular, rectangular and square cross section

3.7.2 Theory:

An industrial piping system may involve pipes of various diameters and may have bends, valves etc., and may not have circular cross sections. Pipe flow will typically involve two important types of energy losses- Major losses and Minor losses. Major losses may be attributed to frictional head loss and minor losses are those occurring at inlets and outlests of pipes, bends, contractions, expansions etc., For turbulent flow in a duct of non-circular cross-section, the hydraulic mean diameter may be used in place of the pipe diameter and the formulae for circular pipes can then be applied without introducing a large error. This method of approach is entirely empirical. To calculate the frictional head loss non-circular pipes the method must be adapted to use the Hydraulic Diameter instead of the internal

	dimensions of the pipe. Hydraulic Diameter = 4 x cross sectional area of flow / wetted perimeter.
3.7.3	**Experimental Set up / Line diagram:***(Students to observe the set-up and draw the line/piping diagram. Symbol list may be referred to prepare the line diagram)*
3.7.4	**Procedure:** ● Connect the manometer between two tapings of the pipe fittings. ● Remove any entrapped air bubbles from the manometer. ● Allow the water to flow through the pipes regulating the flow with gate valve. ● Once manometer readings are steady, note the manometer readings. ● Collect the water for a known amount of time for the adjusted flow rate. ● Repeat the procedure for different flow rates.
3.7.5	Observations: Rectangular Pipe:

Trial No.	Manometer Reading		Flow Rate	
	h_1(cm)	h_2(cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				
4				
5				
6				
7				
8				

9				
10				

Square Pipe:

Trial No.	Manometer Reading		Flow Rate	
	h_1(cm)	h_2(cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Trianglular Pipe:

Trial No.	Manometer Reading		Flow Rate	
	h_1(cm)	h_2(cm)	Volume of water collected (Litre)	Time Taken(s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

3.7.6 **Formulae:**

Specifications:

Length of the pipe $L = 1.5\,m$

Area of the collecting tank $A = 0.125 m^2$

Pipe cross-section:

Sl No	Type of Pipe	Dimensions
1	Triangular	Base $b = 0.025\,m$

		Height $h = 0.025\ m$
2	Rectangular	Length $l = 0.025\ m$ Breadth $w = 0.025\ m$
3	Square	$s = 0.025\ m$

Density of CCl_4 /Mercury at room temperature, ρ_m = _____ kg/m^3

Density of Water at room temperature, ρ = _____ kg/m^3

Viscosity of water, μ = _____ Ns/m^2

Formulae:

Fluid head loss across the pipe:

$R_m = h_1 + h_2\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Flow rate:

$Q = \frac{Area\ of\ the\ tank \times rise\ in\ the\ level}{time}\ \left(\frac{m^3}{s}\right)$

Cross sectional area for Triangular Pipe $A_t = \frac{1}{2} \times b \times h\ (m^2)$

Cross sectional area for Rectangular Pipe $A_R = l \times w\ (m^2)$

Cross sectional area for Square Pipe $A_s = s^2\ (m^2)$

Velocity in triangular pipe , $v = \frac{Q}{A_t}\ \left(\frac{m}{s}\right)$

Velocity in rectangular pipe , $v = \frac{Q}{A_R}\ \left(\frac{m}{s}\right)$

Velocity in square pipe , $v = \frac{Q}{A_s}\ \left(\frac{m}{s}\right)$

Friction factor:

$f = \frac{2g\Delta H}{4Lv^2}$

Calculation of hydraulic mean diameter:

For Triangular Pipe:

$$D_h = \frac{4 \times A_t}{Wetted\ Perimeter}$$

Wetted Perimeter = Sum of all the sides of the triangle

For rectangular Pipe:

$$D_h = \frac{2A_R}{l+w}$$

For Square Pipe:

$$D_h = s$$

Reynold's Number:

$$N_{Re} = \frac{D_h v \rho}{\mu}$$

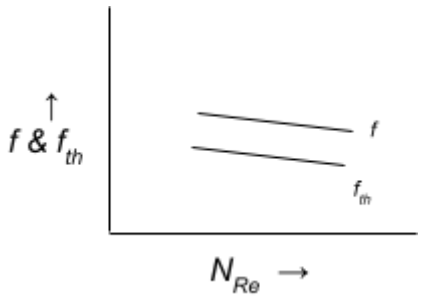
Theoretical value of f:

$f_{th} = \frac{16}{N_{Re}} for\ laminar\ flow,\ N_{Re} < 2100$

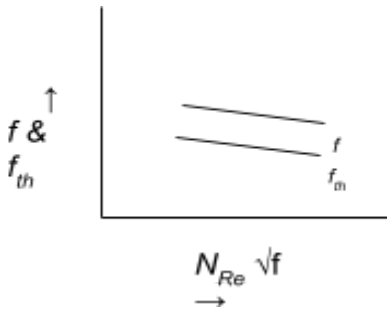
$f_{th} = 0.079 N_{Re}^{-0.25}\ for\ turbulent\ flow\ ,\ N_{Re} > 4000$

Graphs:

$f\ Vs.\ NRe\ and\ f_{th}\ Vs\ N_{Re}\ (log\ -\ log\ graph)$



$b)\ f\ and\ f_{th}\ Vs.\ N_{Re} f\ (log\ -\ log\ graph)$



3.7.7 **Specimen Calculations:**

3.7.8 Results and discussions

3.7.8.1 Results:

For Triangular Pipe:

Trial No	R_m (m)	ΔH (m)	$Q(\frac{m^3}{s})$	$v\ (\frac{m}{s})$	N_{Re}	f	$N_{Re}\sqrt{f}$	f_{th}
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								

For Rectangular Pipe:

Trial No	R_m (m)	ΔH (m)	$Q(\frac{m^3}{s})$	$v\ (\frac{m}{s})$	N_{Re}	f	$N_{Re}\sqrt{f}$	f_{th}
1								
2								
3								
4								
5								

6								
7								
8								
9								
10								

Square Pipe:

Trial No	R_m (m)	ΔH (m)	$Q(\frac{m^3}{s})$	$v\,(\frac{m}{s})$	N_{Re}	f	$N_{Re}\sqrt{f}$	f_{th}
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								

	3.7.8.2	**Discussion:**

3.7.9	**Conclusions:**	
3.7.10	**Assessment**	

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	

		4	Results and discussions	20%	8		
			Total		40		

Signature of the student **Signature of the faculty with date**

3.8 FLOW THROUGH HELICAL COIL

3.8.0 Nomenclature:

μ	Viscosity of fluid flowing through the pipe	Ns/m^2
ρ	Density of fluid flowing through the pipe	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
A_p	Cross sectional area of the pipe	m^2
D_p	Diameter of the pipe	m
D_c	Diameter of the coil	m
f	Friction factor	-
f_{th}	Theoretical friction factor	-
ΔH	Differential head across the pipe	m
ΔH_c	Differential head across the coil	m
ΔH_s	Differential head across the straight pipe	m
L	Length of the coil	m
N_{Re}	Reynolds Number	-
N_{Re_C}	Critical Reynolds Number	m
R_m	Difference in manometer leg levels	m
Q	Volumetric flow rate	m^3/s
v	Velocity of fluid through the pipe	m/s

3.8.1	**Aim:** ● To determine the critical Reynold's number of a fluid flowing through a coil. ● To determine the Deans' number for the fluid flowing through the coil. ● To compare the experimentally determined friction factor values to the theoretical values. ● To compare the pressure, drop in a coil with that in a straight pipe of same length, inside diameter & surface roughness.
3.8.2	**Theory:** Helical coils are used widely in processing industries for cooling and heating applications since the centrifugal forces experienced by the fluid acts to promote contact with the channel wall, thereby tending to insure good contact with the wall, and an enhanced heat transfer. Coils are used in chemical reactors, agitated vessels, and storage tanks to heat and cool process streams ranging from inorganic and organic chemicals to dairy products. When a fluid flows in a curved structure, there exists a secondary flow due to the action of centrifugal force. Such a secondary flow results in high frictional loss in curved pipes than that in straight pipes under similar conditions, such as flow rate, temperature, pressure. For single phase flow in a coil or curved pipe a secondary flow pattern is established in the pipe which changes the fluid behavior and hence friction factor. One effect is the stabilization of fluid flow, which results in an increase in the Reynolds number at which the flow enters the transition from laminar flow to turbulent flow. Critical Reynolds number marks this transition and is given as $$(N_{Re})_c = 2100\left(1+12\sqrt{\frac{D}{D_c}}\right)$$ Reynold's number below Critical Reynolds number the fluid is in the laminar flow regime, above this the fluid is in the transitional or turbulent flow regime. $$(N_{Re})_c = 2100\left(1+12\sqrt{\frac{D}{D_c}}\right)$$

	To determine the friction factor in the coil the Dean number is required and is given as: $De = Re\sqrt{\frac{D}{R}}$ Friction factor may be calculated using different correlations based on the value of Deans number.
3.8.3	**Experimental Set up / Line diagram:** *(Students to observe the set-up and draw the line/piping diagram. Symbol list may be referred to prepare the line diagram)*

3.8.4 Procedure:

- Connect the manometer between two tapings of the pipe fittings.
- Remove any entrapped air bubbles from the manometer.
- Allow the water to flow through the helical coil regulating the flow with gate valve.
- Once manometer readings are steady, note the manometer readings.
- Collect the water for a known amount of time for the adjusted flow rate.
- Repeat the procedure for different flow rates.

3.8.5 Observations:

Trial No.	Manometer Reading		Flow Rate	
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

3.8.6 Formulae:

Specifications:

Length of the pipe $L = 5.9\ m$

Diameter of the pipe, $D_p = 0.006\ m$

Diameter of the coil, $D_c = 0.55\ m/0.45\ m/0.30\ m$

Density of CCl_4 /Mercury at room temperature, ρ_m = _____ kg/m^3

Density of Water at room temperature, ρ = _____ kg/m^3

Viscosity of water, μ = _____ Ns/m^2

Formulae:

Fluid head loss across the pipe:

$$R_m = LHS - RHS\ (m)\Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$$

Flow rate:

$$Q = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$$

Velocity:

$$A_P = \frac{\pi(D_p)^2}{4}\ (m^2)$$

$$v = \frac{Q}{A_p}\ \left(\frac{m}{s}\right)$$

Reynolds Number:

$$N_{Re} = \frac{D_p v \rho}{\mu}$$

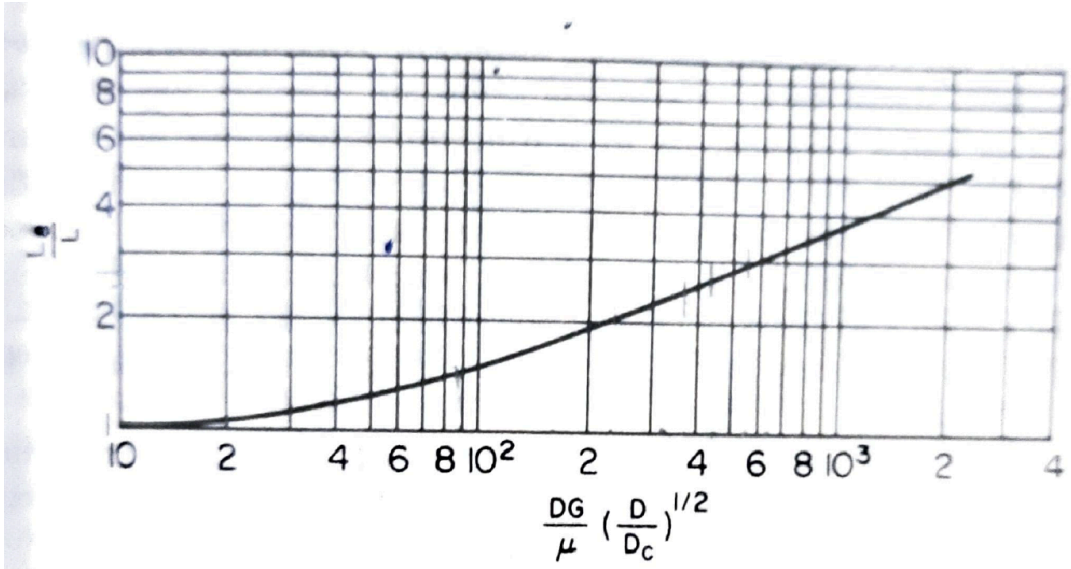
Critical Reynolds number:

$$N_{Re_c} = 2100\left(1 + 12\sqrt{\frac{D_p}{D_c}}\right)$$

Dean's number N_{De}:

$$N_{De} = N_{Re}\sqrt{\frac{D_p}{D_c}}$$

If Dean's number > 10, find the equivalent length from the graph below:



If Dean's number is less than or equal to 10, $L_e = L$

Fanning friction factor:

$$f = \frac{gD_p \Delta H}{2L_e v^2}$$

Theoretical value of f for Helical coil:

$$f_{th_c} = 0.08\left(N_{Re}\right)^{-0.25} + 0.01\left(\frac{D_p}{D_c}\right)^{0.5}$$

Pressure drop in Helical coils:

$$\Delta H_c = \frac{2f_{th}L_e v^2}{gD_p}$$

Theoretical value of f for straight pipes:

$f_{th} = \frac{16}{N_{Re}} for\ laminar\ flow,\ N_{Re} < 2100$

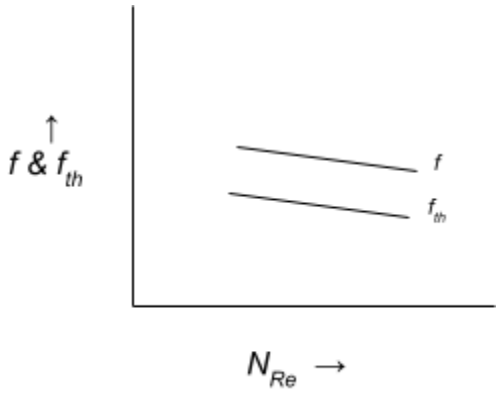
$f_{th} = 0.079N_{Re}^{-0.25}\ for\ turbulent\ flow\ ,\ N_{Re} > 4000$

Pressure drop in Helical coils:

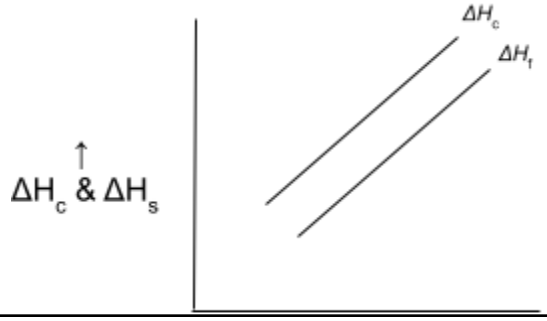
$$\Delta H_s = \frac{2f_{th}L_e v^2}{gD_p}$$

Graphs:

$f\ Vs.\ NRe\ and\ f_{th}\ Vs\ N_{Re}\ (log\ -\ log\ graph)$



ΔH_C v/s $Q\ and\ \Delta H_s$ v/s $Q\ (log\ -\ log\ graph)$



Momentum Transfer

3.8.7	**Specimen Calculations:**

3.8.8 Results and discussions

3.8.8.1 Results:

Sl No	R_m (m)	ΔH (m)	Q $\left(\frac{m^3}{s}\right)$	v $(\frac{m}{s})$	N_{Re}	N_{Po}	f_c	f_{th_c}	ΔH_s (m)	ΔH_c (m)
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										

3.8.8.2 Discussion:

3.8.9	**Conclusions:**	
3.8.10	**Assessment:**	

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**

3.9 FLOW THROUGH PACKED BED

3.9.0 Nomenclature:

μ	Viscosity of fluid flowing through the pipe	Ns/m^2
ρ	Density of fluid flowing through the pipe	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
ϕ_s	Sphericity of the particle	-
ϵ	Porosity of the bed	-
A_b	Cross sectional area of the bed	m^2
D_p	Diameter of the packing particle	m
f	Friction factor	-
f_{th}	Theoretical friction factor	-
f_p	friction factor for a packed bed	-
ΔH	Differential head across the bed	m
L	Length of the bed	m
N_{Re_P}	Particle Reynolds Number	
R_m	Difference in manometer leg levels	m
Q	Volumetric flow rate	m^3/s
v	Superficial velocity of fluid through the bed	m/s
ΔP	Pressure drop across the bed	N/m^2

3.9.1	**Aim:** ● To verify the relationship between the velocity of the fluid and pressure drop per unit length of packing. ● To determine friction factor experimentally ● To compare the values of friction factor obtained experimentally with the theoretical value. ● To verify Ergun's equation.
3.9.2	**Theory:** Chemical Engineering operations commonly involve the use of packed beds. These offer a large surface area of contact between a liquid and a gas, or a solid and a gas or liquid is obtained for achieving rapid mass and heat transfer and for chemical reactions. Packed bed is a cylindrical column with certain packing material. The packing can be randomly filled with small objects like Raschig rings or could be of structured type. Several chemical engineering unit operations such as absorption, adsorption, distillation and extraction are carried out in packed columns. Packed columns are also used for heterogeneous catalytic reactions. Flow through a packed bed can be regarded as fluid flow past a number of submerged objects. When there is no flow through the packed bed, the net gravitational force (including buoyancy) acts downward. When flow begins upward, friction forces act upward and counterbalance the net gravitational force. The frictional force can be expressed in terms of a friction factor. This leads to equations describing the flow of a fluid past a collection of particles. From a fluid mechanical perspective, the most important issue is that of the pressure drop required for the liquid or the gas to flow through the column at a specified flow rate. The pressure losses accompanying the flow of fluids through packed columns are caused by simultaneous kinetic and viscous energy losses. The essential factors determining the energy loss, i.e., pressure drop, in packed beds are: ● Rate of fluid flow ● Viscosity and density of the fluid ● Closeness and orientation of packing ● Size shape and surface of the particles The first two variables concern the fluid, while the last two the solids. To calculate the pressure losses, we rely on a friction factor correlation attributed to Ergun. The frictional force can be expressed in terms of a friction factor. This leads to equations describing the flow of a fluid past a collection of particles. There are several approaches to treating fluid flow through packed beds. The most successful of these is the Ergun Equation, which describes flow in both the laminar and turbulent regimes. This method treats the packed column as a compact irregular bundle of tubes. We assume that velocity, particle diameter and void

fraction behaves as a bulk behavior and hence we can use an average value. The Ergun equation is

$$f_p = \frac{150(1-\epsilon)}{\frac{\phi_s D_p V_0 \rho}{\mu}} + 1.75$$ ---------(1)

$$f_p = \frac{\Delta P \phi_s D_p \epsilon^3}{\rho V_0^2 L(1-\epsilon)}$$ ---------(2)

Ergun equation was obtained by fitting the data for spheres, cylinders and crushed solids such as coke and sand. For Raschig rings and Berl saddles, which have porosities of 0.55 to 0.75, Ergun equation predicts pressure drops lower than those found experimentally. It also does not apply well to other tower packings of high surface area and high porosity.

It is seen from the formula that when the flow is high, the turbulent part of the formula appears; likewise, when the flow is low, the laminar part of the formula comes out. Thus, the Ergun equation can be used for any type of flow.

The Ergun equation tells us the pressure drop along the length of the packed bed given some fluid velocity. It also tells us that the pressure drop depends on the packing size, length of bed, fluid viscosity and fluid density.

At low Reynolds numbers, the quantity 1.75 of Eq. (1) is negligible in comparison with Reynolds number term. This implies that the viscous forces control and that inertial forces are unimportant. Then we have

$$\frac{\Delta P \phi_s^2 D_p^2 \epsilon^2}{\rho V_0 L \mu (1-\epsilon)^2} = 150$$

This is called the Kozeny-Carman equation and is a laminar flow equation, to be used when Reynolds number is less than about 1. For a given system it indicates that the pressure drop is directly proportional to the flow rate.

For large Reynolds numbers, above about 1000, the first term on the right-hand side of Eq. (1) fades out as viscous forces become negligible and inertial forces control, then

$$\frac{\Delta P \phi_s D_p \epsilon^3}{\rho V_0^2 L(1-\epsilon)} = 1.75$$

This is called Blake Plummer equation. For large Reynolds numbers, above about 1000 pressure drop is proportional to square of velocity.

3.9.3 **Experimental Set up / Line diagram:**

(Students to observe the set-up and draw the line/piping diagram. Symbol list may be referred to prepare the line diagram)

3.9.4 **Procedure:**

- Connect the manometer between two tapings of the pipe fittings.
- Remove any entrapped air bubbles from the manometer.
- Allow the water to flow through the packed bed regulating the flow with gate valve.
- Once manometer readings are steady, note the manometer readings.
- Collect the water for a known amount of time for the adjusted flow rate.
- Repeat the procedure for different flow rates.
- Collect the water between packed bed (between two tapings of manometer) for finding void volume.

3.9.5 **Observations:**

Trial No.	Manometer Reading		Flow Rate	
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)
1				
2				
3				
4				
5				

	6				
	7				
	8				
	9				
	10				

3.9.6 **Formulae:**

Specifications:

Length of the packed bed $L = 1.7\ m$

Diameter of the bed, $D = 0.0335\ m$

Shape Factor $\phi_s = 0.3$

Specific surface area of the particle $S_p = 530\ m^2/m^3$

Density of CCl_4 /Mercury at room temperature, ρ_m = _____ kg/m^3

Density of Water at room temperature, ρ = _____ kg/m^3

Viscosity of water, μ = _____ Ns/m^2

Formulae:

Fluid head loss across the pipe:

$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Pressure Drop:

$\Delta P = \rho g \Delta H \quad N/m^2$

Flow rate:

$Q = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$

Area of the bed:

$A_b = \frac{\pi(D)^2}{4}\ (m^2)$

Velocity:

$v = \frac{Q}{A_b}\ \left(\frac{m}{s}\right)$

Diameter of the particle:

$D_p = \frac{6(1-\epsilon)}{\phi_s S_p}\ m$

Particle Reynolds Number:

$$N_{Re_p} = \frac{D_p v \rho}{\mu}$$

Friction factor:

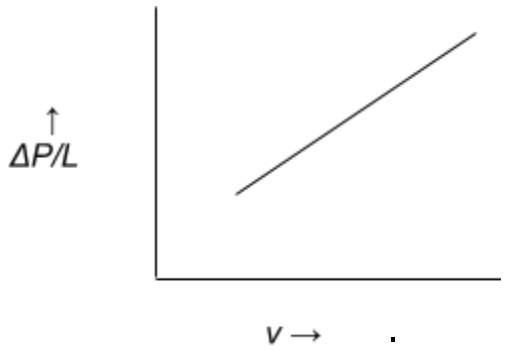
$$f = \left(\frac{\Delta P}{L}\right)\left(\frac{\epsilon^3}{1-\epsilon}\right)\left(\frac{D_p}{\rho v^2}\right)$$

Theoretical Friction factor:

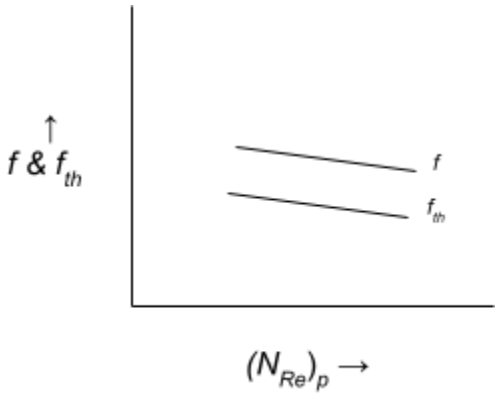
$$f_{th} = \left[\frac{150(1-\epsilon)}{N_{Re_p}}\right] + 1.75$$

Graphs:

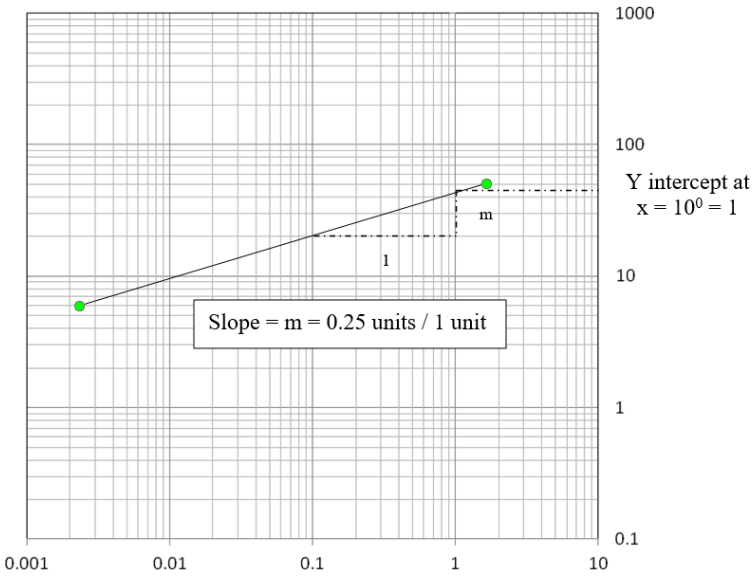
Draw a graph of $\Delta P/L \ v/s \ v$ on ordinary graph. Obtain the value of slope (n)



$f \ Vs. \ N_{Re_p} \ and \ f_{th} \ Vs \ N_{Re_p} \ (log - log \ graph)$



$f \ v/s \ N_{Re_p}$ (log-log graph) (To verify Ergun's equation)



Find the intercept of the line from the above graph an compare it with 1.75.

Find the slope of the line from the graph and compare it with $150(1 - \epsilon)$.

3.9.7 **Specimen Calculations:**

3.9.8 **Results and discussions**

3.9.8.1 **Results:**

Sl No	R_m m	ΔH m	ΔP $N/m2$	Q m^3/s	v m/s	N_{Re_P}	f	f_{th}	$\Delta P/L$
1									
2									
3									
4									
5									

6									
7									
8									
9									
10									

	3.9.8.2	**Discussion:**
3.9.9	**Conclusions:**	

3.9.10	**Assessment:**

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**

3.10 FLOW THROUGH FLUIDIZED BED

3.10.0 Nomenclature:

μ	Viscosity of fluid flowing through the pipe	Ns/m^2
ρ	Density of fluid flowing through the pipe	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
ρ_p	Density of the particle	kg/m^3
ϵ_o	Porosity of fixed bed	-
ϵ	Porosity of fluidized bed	-
A_b	Cross sectional area of the pipe	m^2
D_b	Diameter of the pipe	m
D_p	Diameter of the particle	m
ΔH	Differential head across the bed	m
L	Length of the fluidized bed	m
L_o	Length of the fixed bed	m
N_{Re_p}	Particle Reynolds Number	-
ΔP	Pressure drop across the bed	N/m^2
R_m	Difference in manometer leg levels	m
Q	Volumetric flow rate	m^3/s
v	Velocity of fluid through the bed	m/s

3.10.1 Aim:

- To determine pressure, drop per unit bed length as a function of superficial velocity of fluidizing medium.
- To calculate the minimum fluidization velocity graphically and compare with the standard value.

3.10.2 Theory:

Fluidization is one of the methods available for contacting granular solids with fluids. The fluidization technique was first started by the petroleum refineries to find a better catalytic cracking process than the fixed bed. The main advantages of fluidized bed are greater interfacial surface area of contact, high rates of heat transfer, avoidance of hot spots & ease in solid handling. Fluidized beds also find application in chemical reactions.When a fluid passes upward through a bed of solids, there will be a certain pressure drop across the bed required to maintain the fluid flow. Depending upon the bed geometry, fluid velocity &

	particle characteristics the following phenomena occur with gradual increase in fluid velocity. At low fluid velocities, there is a pressure drop across the bed but the particles are stationery & the flow of fluid is through a fixed bed. As the bed velocity is gradually increased, a certain velocity is reached when the bed starts expanding. At this point the pressure drop across the bed equals the mass per unit area of the bed. This point is known as incipient Fluidization. With increase in fluid velocity the particles separate still further from one another, the bed continues to expand, the porosity increases, but the pressure difference does not change.
3.10.3	**Experimental Set up / Line diagram:** *(Students to observe the set-up and draw the line/piping diagram. Symbol list may be referred to prepare the line diagram)*
3.10.4	**Procedure:** ● Connect the manometer between two tapings of the pipe fittings. ● Remove any entrapped air bubbles from the manometer. ● Allow the water to flow through the packed bed regulating the flow with gate valve. ● Once manometer readings are steady, note the manometer readings. ● Collect the water for a known amount of time for the adjusted flow rate. ● Repeat the procedure for different flow rates. ● Collect the water between packed bed (between two tapings of manometer) for finding void volume.

3.10.5 **Observations:**

Trial . No	Manometer Reading		Flow Rate		Bed Length L(m)
	LHS (cm)	RHS (cm)	Amount of water collected (Litre)	Time Taken (s)	
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

3.10.6 **Formulae:**

Specifications:

Length of the fixed bed $L_o = 1.12\ m$

Diameter of the bed, $D = 0.045\ m$

Diameter of the particle, $D = 0.004\ m$

Density of the particle, $\rho_p = 2600\ kg/m^3$

Sphericity $\phi_s = 1$

Density of CCl_4 /Mercury at room temperature, ρ_m = _____ kg/m^3

Density of Water at room temperature, ρ = _____ kg/m^3

Viscosity of water, μ = _____ Ns/m^2

Porosity of the fixed bed, $\epsilon_o = 0.36$

Formulae:

Fluid head loss across the pipe:

$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Pressure Drop :

$\Delta P = \rho g \Delta H \quad N/m^2$

Flow rate:

$$Q = \frac{Volume\ of\ water\ collected}{time} \quad \left(\frac{m^3}{s}\right)$$

Area of the bed:

$$A_b = \frac{\pi(D)^2}{4} \quad (m^2)$$

Velocity:

$$v = \frac{Q}{A_b} \quad \left(\frac{m}{s}\right)$$

Porosity of the bed :

$$\epsilon = \left(1 - \left(\frac{L_o}{L}\right)(1 - \epsilon_o)\right)$$

Particle Reynolds Number:

$$N_{Re_p} = \frac{D_p v \rho}{\mu}$$

Minimum fluidization Velocity:

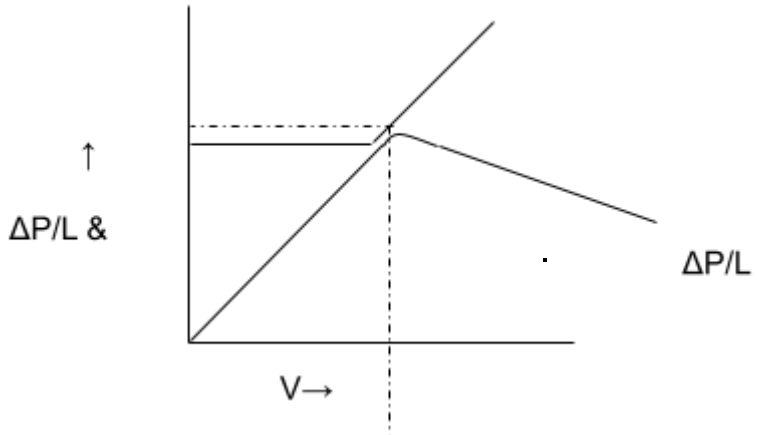
$$v_o = \left[\phi_s D_p g \frac{(\rho - \rho_p)\epsilon_o^3}{1.75\,\rho}\right]^{0.5} m/s$$

Free settling velocity:

$$v_f = 1.75\left[D_p g \frac{\rho_p - \rho}{\rho}\right]^{0.5} m/s$$

Graphs:

Draw a graph of $\frac{\Delta P}{L}\ v/s\ v\ and\ \epsilon\ v/s\ v$ on log-log graph



3.10.7	**Specimen Calculations:**

3.10.8 Results and discussions

3.10.8.1 Results:

Sl No	R_m $_m$	ΔH m	ΔP N/m^2	$^{\Delta P/L}$	Q m^3/s	v m/s		N_{Re_P}
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								

3.10.8.2 Discussion:

3.10.9	**Conclusions:**	
3.10.10	**Assessment:**	

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	

			Total	40		

Signature of the student **Signature of the faculty with date**

3.11 FLOW THROUGH CENTRIFUGAL PUMP

3.11.0 Nomenclature:

μ	Viscosity of fluid	Ns/m^2
ρ	Density of fluid	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
η	Efficiency of the pump	-
ΔH	Differential head across the pipe	m
OHP	Output power of the pump	hp
IHP	Input power of the pump	hp
n	No of revolutions	-
R_m	Difference in manometer leg levels	m
Q	Volumetric flow rate	m^3/s
t	Time taken for n revolutions	s

3.11.1 Aim:

- To determine the efficiency of pump at different flow rates
- To study the performances by plotting the characteristic curves.
- To determine the optimum condition at which the pump is to be operated.

3.11.2 Theory:

Centrifugal pumps are most often used in water and wastewater systems.

They are hydraulic machines that convert mechanical energy into pressure energy. It works on the principle of forced vertex flow, which means that certain mass of liquid is rotated by an external torque. Centrifugal pumps have several advantages over other types of pumps, including:

- Simplicity of construction – no valves, no piston rings, etc.;
- High efficiency;

- Ability to operate against a variable head;
- Suitable for being driven from high-speed prime movers such as turbines, electric motors, internal combustion engines etc.; and
- Continuous discharge.

A centrifugal pump consists of a rotating shaft that is connected to an impeller, which is usually comprised of curved blades. The impeller rotates within its casing and sucks the fluid through the eye of the casing (point 1 in Figure 3.11.1). The fluid's kinetic energy increases due to the energy added by the impeller and enters the discharge end of the casing that has an expanding area (point 2 in Figure 10.1). The pressure within the fluid increases accordingly.

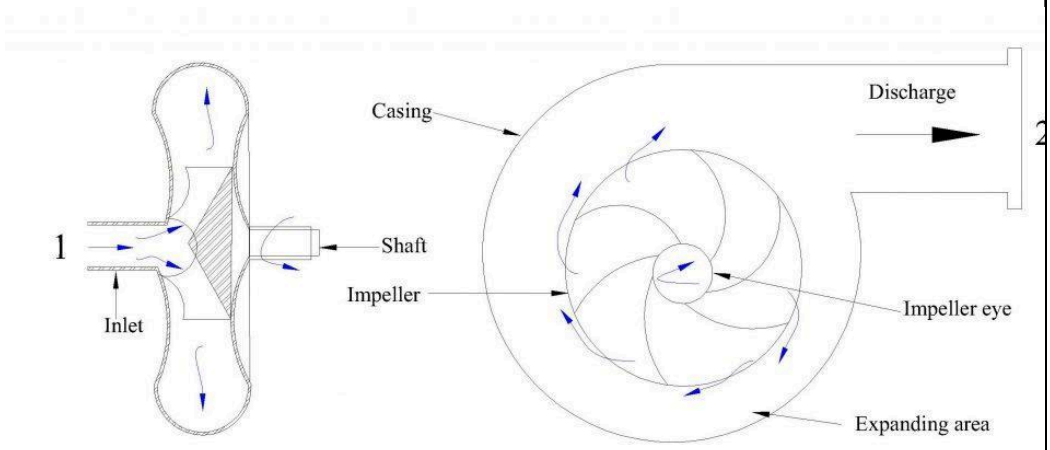


Figure 3.11.1: Schematic of a typical centrifugal pump

The performance of a centrifugal pump is presented as characteristic curves in Figure 3.11.2, and is comprised of the following:

- Pumping head versus discharge,
- Input power versus discharge, and
- Efficiency versus discharge.

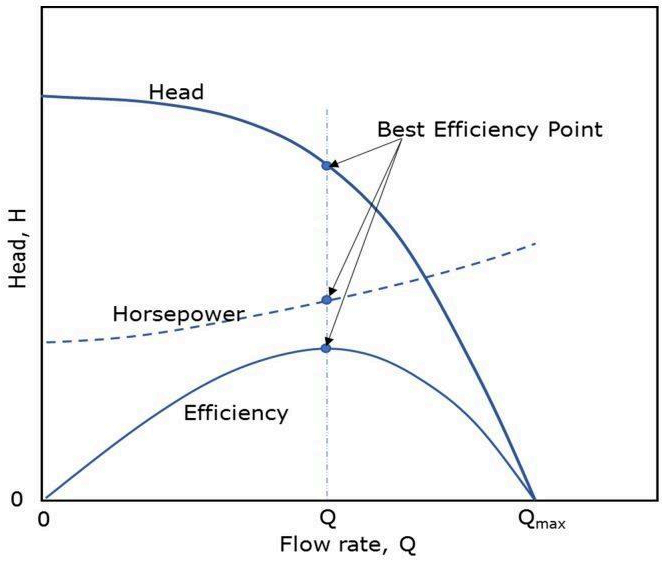


Figure 3.11.2: Typical centrifugal pump performance curves

3.11.3 **Experimental Set up / Line diagram:**

(Students to observe the set-up and draw the line/piping diagram. Symbol list may be referred to prepare the line diagram)

3.11.4 **Procedure:**

- Before starting the experiment check if the valves in the line are opened.
- Ensure that the pump is primed by filling the pump with water.
- The motor is then switched on.
- The flow rate is regulated using the globe valve.
- The flow is allowed to attain steady state before taking the reading.
- The following observations are recorded.
  - Manometer reading.
  - Flow rate by collecting water for a known amount of time.
  - Time taken for 5 revolutions on the power meter.
- The above experiment is repeated at least six times starting from maximum to minimum flow rate maintaining steady state in each run.

3.11.5 **Observations:**

Trial No.	Manometer Reading		Flow Rate		Time taken for 5 revolutions (s)
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)	
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

3.11.6 **Formulae:**

Specifications:
Height from datum point where pressure gauge is connected in delivery line
$Z_b = 0.285\ m$

Area of the tank = $735\times10^{-4}\ m^2$

Density of the manometric fluid $CCl_4/Mercury$ ρ_m = _____ kg/m^3

Density of Water at room temperature, ρ = _____ kg/m^3

Viscosity of water, μ = _____ Ns/m^2

Formulae:

Fluid head loss across the pipe:

$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Fluid Head:

$H = \Delta H + Z_b\ (m)$

Pressure Drop :

$\Delta P = \rho g \Delta H \quad N/m^2$

Flow rate:

$Q = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$

Output power of the pump :

$OHP = \frac{Q\ \rho\ g\ H}{746}\ (hp)$

Input power of the pump :

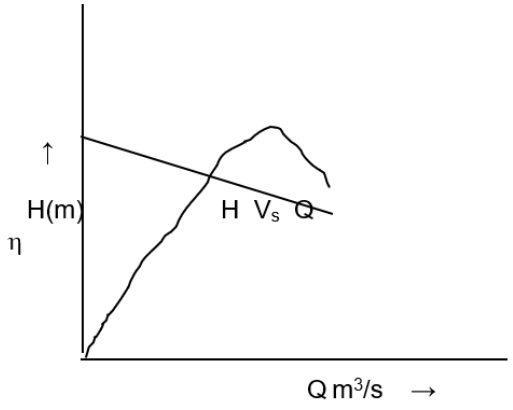
$IHP = \frac{n\times3600\times10^3}{t\times400\times746}\ (hp)$

Efficiency of the pump:

$\eta = \left(\frac{OHP}{IHP}\right)\times100$

Graphs:

a) H Vs. Q b) η Vs. Q



The optimum condition at which the pump is to be operated is the point where the head curve cuts the ordinate through the point of maximum efficiency.

3.11.7	**Specimen Calculations:**

3.11.8 Results and discussions

3.11.8.1 Results:

Sl No	R_m $_m$	H m	Q m^3/s	OHP hp	IHP hp	
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

	3.11.8.2	**Discussion:**
3.11.9	**Conclusions:**	
3.11.10	**Assessment:**	

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**

3.12 FLOW THROUGH AIR LIFT PUMP

3.12.0 Nomenclature:

ρ	Density of air	kg/m^3
ρ_m	Density of manometric fluid	kg/m^3
A_1	Cross sectional area of the pipe	m^2
A_2	Cross sectional area of the orifice	m^2
c_d	Coefficient of discharge of the orifice	-
d_1	Diameter of the pipe	m
d_2	Diameter of the orifice	m
IHP	Input power to the compressor	hp
ΔH	Differential head of air	m
L	Length of the pipe	m
OHP	Output power to the compressor	hp
R_m	Difference in manometer leg levels	m
Q	Volumetric flow rate of water	m^3/s

	Q_a | Volumetric flow rate of water | m^3/s
3.12.1	**Aim:** ● To determine effect of between air flow rate and on water discharge in an air lift pump. ● To determine the efficiency of the air lift pump.
3.12.2	**Theory:** Pumps facilitate fluid transport from one point to another by virtue of a pressure difference which it creates utilizing external power. Pumps can be divided into three categories: Positive displacement, Rotodynamic and Dynamic. Airlift pumps are of dynamic type. Figure 1 shows the schematic of arrangement of an airlift pump. An airlift pump consists of a cylindrical conduit which is called a riser. The only energy required is that of the pressure energy of compressed air. Air from a compressor/blower is injected at the lower end of the riser tube. This tube is initially submerged partially in the liquid to be pumped. Buoyancy causes the air to rise upwards and carries with it a substantial amount of entrained liquid mass causing ascent of the liquid. This is a classic example of a multiphase (two-phase) flow system. Airlift pumps are used in sea-bed mining sector for the extraction of mineral like manganese. Airlift pumps are also useful for pumping of corrosive chemicals, which would otherwise potentially wear off the impellers and pistons of conventional pumping devices. Aeration of ponds in aquaculture is also a typical application area of airlift pumps. A very specialized application is for archaeological drills, especially at sea. Airlift pumps have exceptional ability to pump dirty fluids. Another huge advantage is the absence of any priming requirements due to the fact that cavitation does not occur in airlift pumps. The foremost strength of an airlift pump is its simplicity. No moving parts imply no damage, no critical maintenance, no special inventory requirements and almost nil overhauling costs. Reliability is high and failure probability is low. Only driving force required is compressed air. Low efficiency is the foremost disadvantage. Airlift pump components have non-standard design. A major drawback is the impossibility of horizontal operation. Most of the times, much less mass of liquid can be lifted per mass of air injected. Thus, compression cost is high. Airlift pumps offer more or less constant discharge, with negligible flow control and variation. Discharge is periodic and cause oscillation and vibration

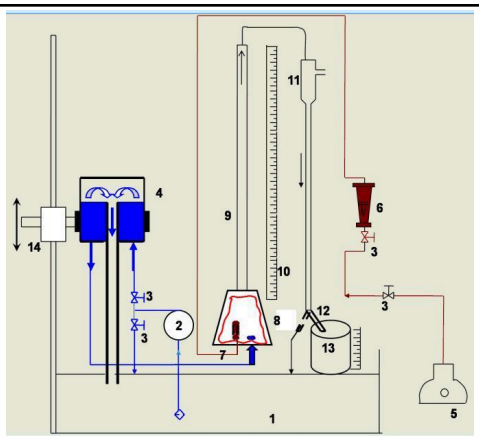


Fig.1: A Schematic of the Airlift Pump Experimental Setup.

(1. Sump. 2. Pump. 3.Control valves. 4. Vessel for level adjustment 5. Compressor. 6. Rotameter. 7.Air injector. 8. Mixing cone. 9. Central core(educator).10. Ruler. 11. Separator. 12. Toggle. 13. Measuring bucket. 14. Level adjustment screw.)

3.12.3 **Experimental Set up / Line diagram:**

(Students to observe the set-up and draw the line/piping diagram. Symbol list may be referred to prepare the line diagram)

3.12.4 **Procedure:**

- The sump tank is filled with clean water.
- Ensure delivery valve is closed and manometer valve is open.
- Ensure safety of electrical connections to the system.
- Switch the compressor on and open the delivery valve.
- Adjust the flow rate of air using control valve of the compressor.
- Note the differential head in the manometer.
- Note the reading from the collecting tank for known time to get the flow rate of water collected.

- The trials are repeated by hanging the air flow rate.

3.12.5 **Observations:**

Trial No.	Manometer Reading		Flow Rate		Time taken for n revolutions (s)
	LHS (cm)	RHS (cm)	Volume of water collected (Litre)	Time Taken (s)	
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

3.12.6 **Formulae:**

Specifications:

Density of air $\rho = _____ \; kg/m^3$

Density of manometric fluid $\rho_m = _____ \; kg/m^3$

Diameter of the air pipe $d_1 = 0.014\, m$

Diameter of the orifice $d_2 = 0.008\, m$

Coefficient of discharge of orifice $c_d = 0.623$

Energy meter constant $EMC = 3200\, revolutions/kWh$

Formulae:

Air Head :

$R_m = LHS - RHS\ (m) \Delta H = \frac{R_m(\rho_m - \rho)}{\rho}\ (m)$

Area of air pipe

$A_1 = \frac{\pi d_1^4}{4}\ (m^2)$

	Area of orifice $$A_1 = \frac{\pi d_2^4}{4}\ (m^2)$$ Volumetric Flow rate of water : $$Q_a = \frac{c_d \times A_1 \times A_2 \times \sqrt{2g\Delta H}}{\sqrt{A_1^2 - A_2^2}}$$ Volumetric Flow rate of water : $Q = \frac{Volume\ of\ water\ collected}{time}\ \left(\frac{m^3}{s}\right)$ Output power of the pump : $OHP = \frac{(Q_a \rho g \Delta H)}{746}\ \ (hp)$ Input power of the pump : $IHP = \frac{n \times 3600}{t \times EMC \times 746}\ \ (hp)$ Efficiency of the pump : $$\eta = \frac{OHP}{IHP} \times 100$$ Draw a graph of $Q\ v/s\ Q_a$ 
3.12.7	**Specimen Calculations:**

3.12.8 **Results and discussions**

3.12.8.1 **Results:**

Sl No	R_m $_m$	ΔH m	Q m^3/s	Q_a m^3/s	OHP hp	IHP hp	
1							
2							
3							
4							
5							

	6							
	7							
	8							
	9							
	10							

	3.12.8.2	**Discussion:**
3.12.9	**Conclusions:**	

3.12.10	**Assessment:**

Sl. No	Assessment Criteria	Percentage of marks	Max. Marks	Marks obtained
1	Preparedness	20%	8	
2	Viva Voce	20%	8	
3	Conduction of Experiment	40%	16	
4	Results and discussions	20%	8	
	Total		40	

Signature of the student **Signature of the faculty with date**